



FONTAINE_DECAL User Guide

Automatic Decal Projection System for Fontaine Modification

Understanding setup, daily operations, and engineering effort for further development

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1. Overview

1.1 System Overview

This design automates truck door decal placements through the use of a computer vision-based projection system. The camera automatically detects door geometry and calculates accurate decal positions. The projector then displays decal images directly onto the door, allowing the operator to quickly and consistently apply decals with no manual measurements.

1.2 System Purpose

The purpose of this system is to reduce manual measurement, improve placement accuracy, and increase efficiency in high-volume decal application processes.

1.3 Intended Use

This system is intended for use in controlled shop environments where truck doors can be positioned directly in front of the camera and projector. Operators should use the system to assist with decal placement by following projected alignment guides.

2. System Components

2.1 Hardware

- **Camera – See3CAM CU_135**
 - Captures 13MP high-resolution images of the truck door for processing. The camera provides the input used by the segmentation model to detect door geometry and features.
- **Projector – Viewsonic LS740 HD**
 - Projects decal alignment guides directly onto the truck door. The projected image serves as a visual reference for accurate decal placement.
- **Microcomputer – NVIDIA Jetson Orin Nano**
 - Runs the segmentation model and executes the image processing pipeline. It processes the captured image, computes decal placement, and sends projection output to the projector.
- **Mounting System**
 - Provides a stable and adjustable platform for positioning the camera and projector. Proper alignment of the mounting system ensures consistent system performance and accuracy.

2.2 Software

- **Python – Programming Language**
 - Primary language used to run the system and coordinate all processing tasks.
- **OpenCV – Computer Vision Library**
 - Processes images to extract door contours and compute decal placement geometry.
- **Ultralytics YOLOv8 – Segmentation Model**
 - Detects the truck door in the captured image and generates a segmentation mask.

- **ChArUco – Calibration & Mapping**
 - Used for camera and projector calibration to ensure accurate mapping between image coordinates and real-world positions.
- **RS-232 – Projector Communication**
 - Enables communication between the microcomputer and projector to control projection output.
- **FastAPI & Uvicorn – GUI & Web Server**
 - Hosts the user interface and allows the operator to start the system and monitor progress.

2.3 System Connections

The camera captures images and transmits them to the microcomputer for processing. The microcomputer runs the segmentation and geometry algorithms, then outputs the computed decal placement to the projector. The projector displays the alignment guides onto the truck door for operator use.

3. System Setup

3.1 Powering On Components

1. Power on the NVIDIA Jetson Orin Nano and allow the system to fully boot.
2. Turn on the projector and ensure it is displaying output.
3. Confirm the camera is connected and recognized by the system.
4. Wait for all components to initialize before proceeding.

3.2 Verify Connections

- Verify the projector is receiving output and is properly focused on the door.
- The focus ring on the projector has been marked to make focusing at 8ft repeatable.
- Check that all cables (USB, HDMI/DisplayPort, RS-232, and power) are secure.
 - See Figure 1 below.
- Ensure the user interface loads correctly on another device on the same network.

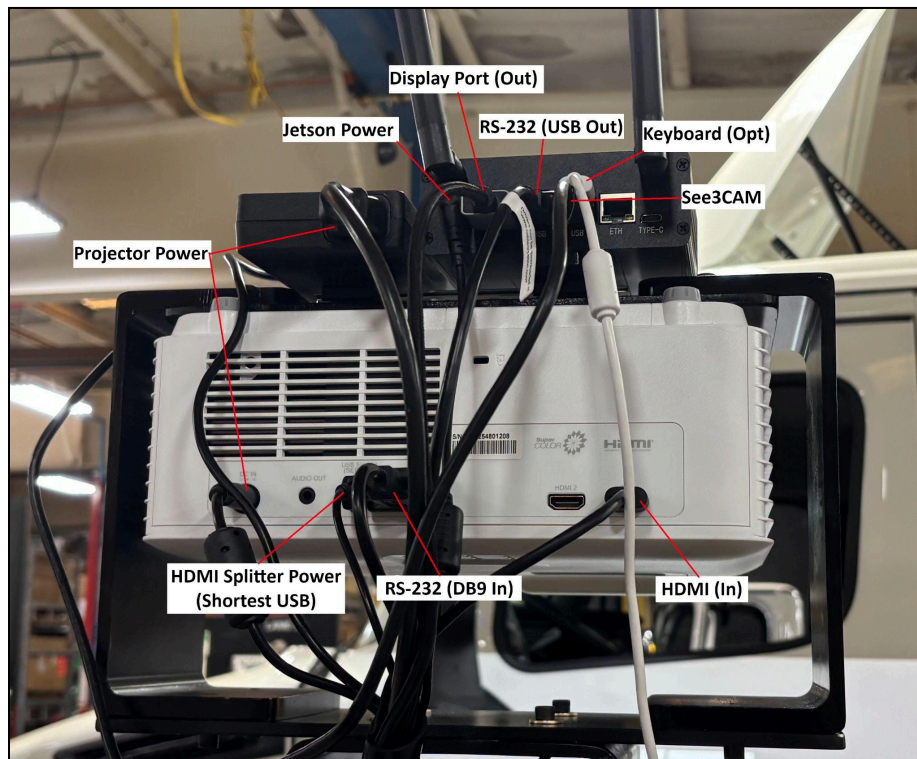


Fig. 1 Sensing Head Wiring Diagram

3.3 Positioning the Truck

1. Park the truck such that the target door will be fully visible within the camera frame.
2. Keep the door closed and free of obstructions.
3. Ensure the entire door boundary is unobscured.

3.4 Positioning the System

1. Place the mobile system approximately 8 ft from the truck door.
2. The system should be facing near-perpendicular to the truck door.
3. Lock all four caster wheels to ensure the system is stable and remains in position.
4. Use the two locking handles on the side of the projector mount to ensure the bottom edge of the projection window is below the bottom edge of the door.
5. Avoid moving the system after alignment unless intentional and necessary.

3.5 Environment Conditions

- Ensure adequate and consistent lighting across the door surface.
- Minimize glare, reflections, or strong shadows on the door.
- Keep the surrounding background as clear as possible.

4. Operating Procedure

4.1 Pre-Run Checklist

Before pressing START in the user interface, verify:

1. Mounting system is stable
2. Camera lens cover is removed
3. System is powered on and initialized
4. Door is straight-on and fully visible
5. No major glare, shadows, or obstructions are present
6. Truck door is centered in the camera view using the PREVIEW CAMERA button (Fig. 2)

4.2 Start the System

Login to the system with user credentials. Press the green START button on the dashboard to begin the main workflow (Fig. 2).

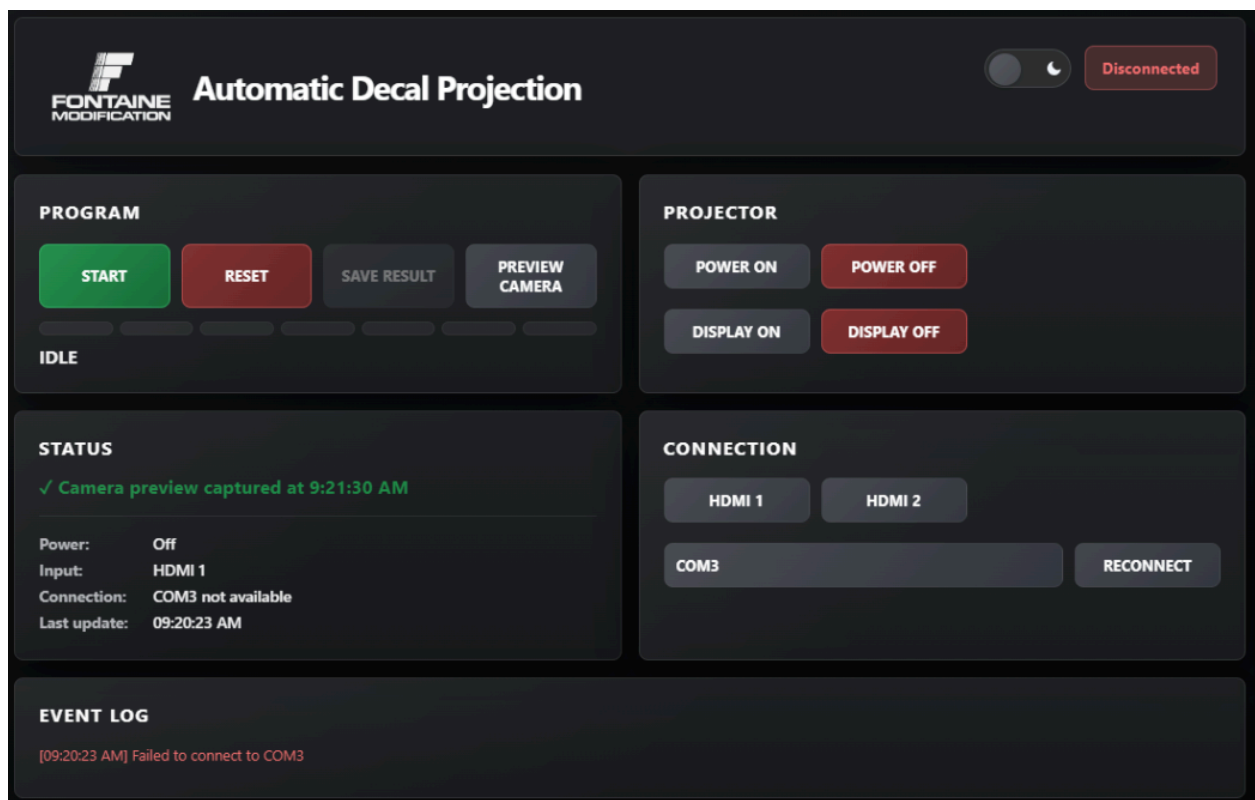


Fig 2. GUI Main Dashboard

4.3 System Processing

The system will automatically:

- Capture two images of the truck door, first with calibration checkerboard, then blank
- Run segmentation to detect door geometry and calculate orientation
- Compute decal placement
- Prepare projection output according to decal layout specification

Monitor the progress bar underneath the START button during this process.

4.3 Decal Projection

The system will automatically project a clean rendering of the decals which have been pre-warped to appear undistorted on the door's surface (Fig. 3). It is possible to display a more detailed version of the layout with centerline annotations for verification purposes. (Refer to Section 5. Engineering Efforts.)



Fig 3. Detailed Projection Example

4.5 Decal Application

- Use the projected images to position and apply decals accurately.
- (Verification only) Follow the projected centerline and markers for alignment.

4.6 User Interface Controls

PROGRAM

- **START** - Begins the main workflow
- **RESET** - Resets progress of the main workflow
- **SAVE RESULT** - Saves images of door after running main workflow
- **PREVIEW CAMERA** - Displays image of the current camera frame (not live)
- **Progress bar** - Text updates on pipeline status

PROJECTOR

- **POWER ON / POWER OFF** - Turns the projector on or off
- **DISPLAY ON / OFF** - Temporarily disables or enables the the projected image

CONNECTION

- **HDMI 1 / HDMI 2** - Toggles between HDMI 1 and HDMI 2
- **/dev/ttyUSB0** - USB serial connection for projector

STATUS

- Displays current status of the system, any raised system errors, and connection info

EVENT LOG

- Displays a timestamped list of recent messages

PROFILE (Top right icon)

- Menu with options to Logout, Force Shutdown, and Restart the system

THEME (Toggle button next to Connectivity Status)

- Toggles between light and dark themes

5. Engineering Efforts

5.1 Use CAD to Find Major Door Plane

This step is necessary for achieving accurate projection mapping and is only required to be done once for each unique truck model. The goal is to choose three points on the door that best approximate the plane that will be projected onto. These same three points must be able to be accurately detected by the custom trained machine vision model. Additionally, to compute an accurate homography four points will be required. Select another distinct door feature that the machine vision model will be able to detect.

The required materials to begin are a 3D CAD model of the truck door, preferably separated from the rest of the cab, and CAD software. The example below will use Autodesk Inventor and the Freightliner Cascadia P4. The exact workflow will vary depending on the specific software and truck model.

Start by placing working points on the three points that will define the door plane and the fourth point for homography calculation. In the case of the driver side door of the Cascadia P4, these points will be the bottom left, bottom right, top right, and mirror mount bottom right (Fig. 4). The work points have been placed near the center of the radius on each corner, as this closely matches where the machine vision model will identify the corner. From these points, use the Plane tool to define a plane from the three corner points (Fig. 5).

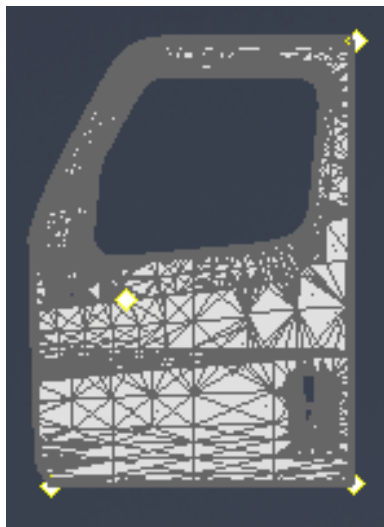


Fig 4. Four Points Selected

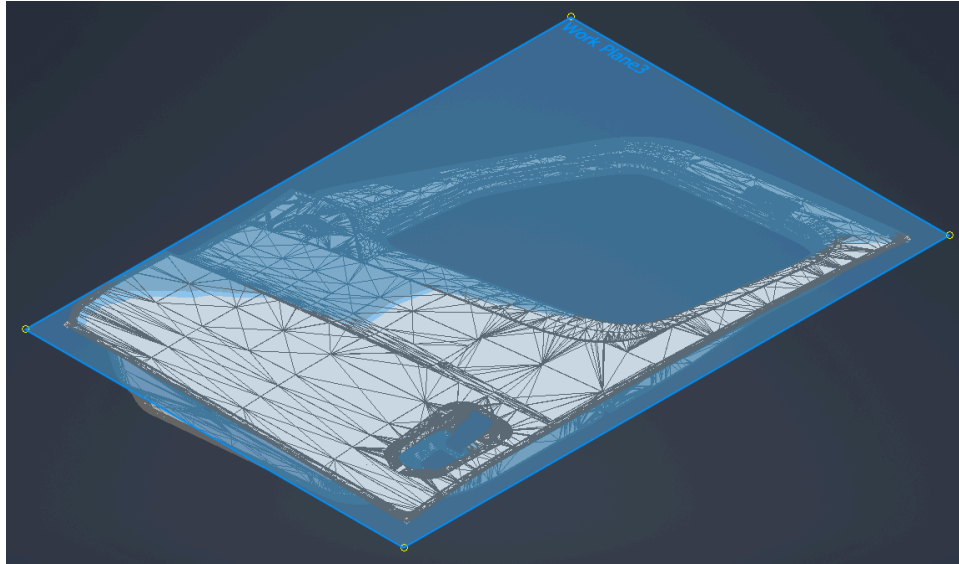


Fig 5. Approximated Door Plane

To approximate the axis that will define the coordinates for decal placement, some auxiliary points are needed. Place a total of three more points, two near the end of the bottom right axis, and another at the end of the region of interest on the right edge of the door (Fig. 6). This is done because the bottom right corner is not a perfect 90° angle, the y-axis for decal placement can become skewed if using the entire right edge. The bottom left corner will be reused for defining the x-axis.



Fig 6. Auxiliary Points for Axis Definition

Use the axis tool and define the x- and y-axis as the line intersecting each pair of auxiliary points (Fig. 7).

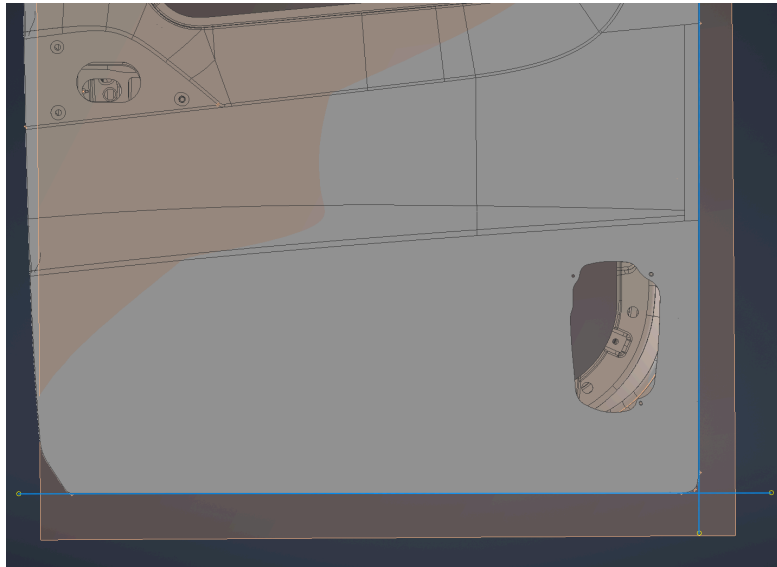


Fig 7. Defined X- and Y-axis

Next, start a 2D Sketch on the new plane. Use the Project Geometry tool to project both defined axes onto the plane, and place a 2D point where they intersect. To properly define the orientation of the User Coordinate System (UCS) that will be placed, a feature on the positive x and y-axis is needed. Place a point on the positive side of each of the projected axes (Fig. 8).

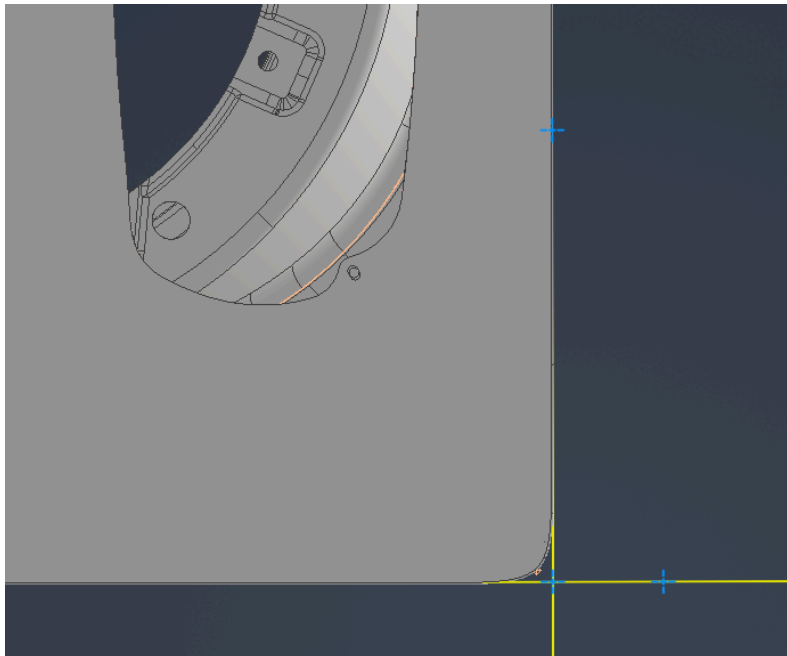


Fig 8. Positive X-Axis Point

After this, the 2D sketch can be finished and the UCS can be placed on the intersection point just drawn. Select the new floating points for the direction of each axis. This will automatically define the positive z-axis to be away from the door (Fig. 9).

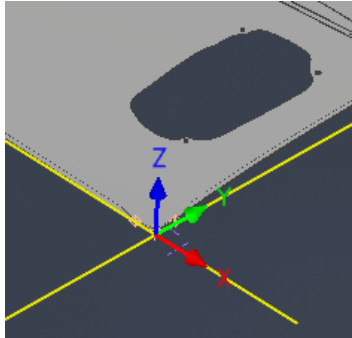


Fig 9. Correct UCS Orientation

With this local coordinate system defined, the dimensions needed to compute decal warping can be measured. Using the measure tool, measure the location of all four projected features in the door plan with respect to the UCS origin (Fig. 10). See Table 1 for the driver side Cascadia P4 door measurements.

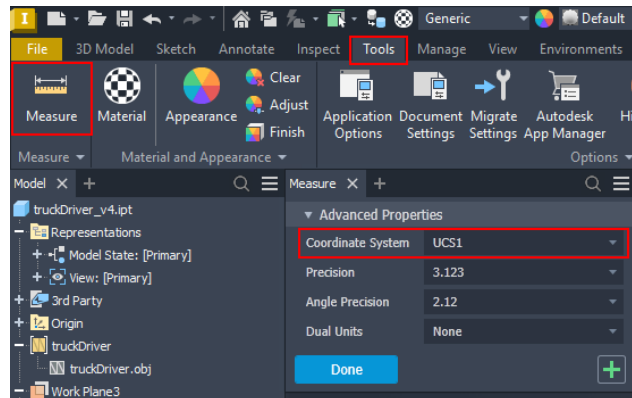


Fig 10. Measurement Configuration

Point	X	Y	Z
Bottom right	-0.328"	0.227"	0.000"
Bottom left	-42.369"	0.000"	0.000"
Mirror	-32.417"	26.358"	-0.6654"
Top right	0.006"	62.565"	0.000"

Table 1. Cascadia P4 Driver's Side Door Measurements

The order of the points is critical, and is defined in the 'DOOR_FEATURE_NAMES' list in */homography/final_projection_pipeline.py* (line 62). The measurements for each point must go in the array that follows in the same order (Fig. 11). Once these points are defined for each unique door model they should not be changed.

```
.venv > Final_Pipeline > homography > final_projection_pipeline.py > DOOR_FEATURE_NAMES
62 DOOR_FEATURE_NAMES = ("bottom_right", "bottom_left", "mirror_mount_right", "top_right")
63 DOOR_FEATURE_POINTS_XY_IN = np.array(
64     [
65         [ 0.000, 0.000],
66         [-43.321, 0.000],
67         [-31.896, 26.161],
68         [ 0.555, 62.299],
69     ],
70     dtype=np.float64,
71 )
```

Fig 11. Store Feature Measurements

5.2 One-Time Calibration

Camera calibration only needs to be performed once for each new camera after the focal length has been set. Capture at least twenty images of the calibration board with the See3CAM before running the calibration script. The images should show the board in a variety of positions and orientations. Figure 12 shows an example of what these images should look like. Save the images to */calibration/raw_images/* and run *camera_calibration.py* from the terminal. This will compute the matrices for the camera's intrinsic and distortion values.

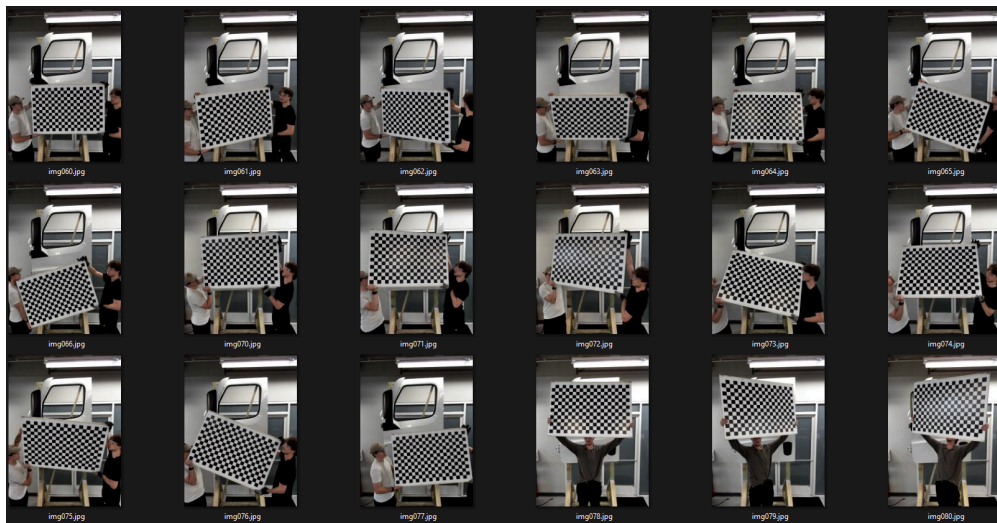


Fig 12. Camera Calibration Images

The results are saved in *results.json* (Fig. 13) and must be copied over to 'CAMERA_MATRIX' and 'DIST_COEFFS' in */homography/final_projection_pipeline.py*, lines 48 and 56 respectively (Fig. 14).

```

1  {
2    "Calibration Date/Time": "04/10/2026 14:06:29",
3    "Camera Matrix (mtx or K)": [
4      [
5        3960.5847485906015,
6        0.0,
7        1526.7108861070315
8      ],
9      [
10       0.0,
11       3969.565061131087,
12       2140.117606363049
13     ],
14     [
15       0.0,
16       0.0,
17       1.0
18     ]
19   ],
20   "Distortion Coefficients (dist)": [
21     [
22       0.11920817196405799,
23       -0.1279512641479141,
24       -0.0003471140112831076,
25       0.001354749232703227,
26       -0.25605877395846227
27     ]
28   ]

```

Fig 13. 'results.json'

```

48  CAMERA_MATRIX = np.array(
49    [
50      [3960.58477, 0.0, 1526.71088],
51      [0.0, 3969.56508, 2140.11761],
52      [0.0, 0.0, 1.0],
53    ],
54    dtype=np.float64,
55  )
56  DIST_COEFFS = np.array([0.11920817, -0.12795116, -0.00034711, 0.00135475, -0.25605914], dtype=np.float64)

```

Fig 14. Camera Intrinsics and Distortion Coefficients Location

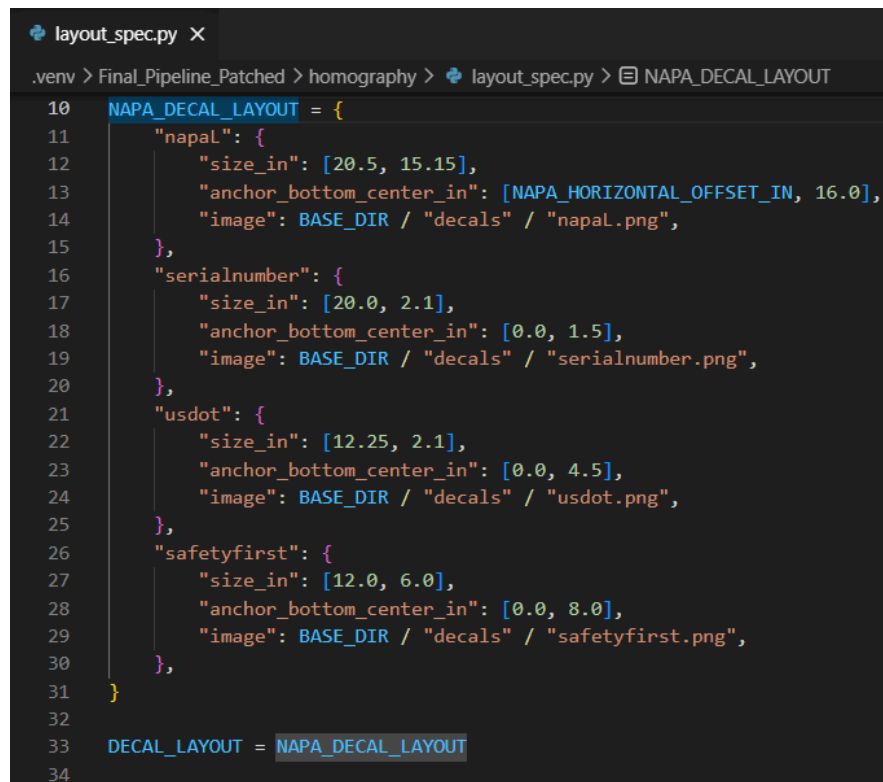
Projector calibration must be performed at the intended projection distance of 8ft. The required materials are the system and a flat white surface large enough for the entire projection window. Project the fullscreen *checkerboard.png* found in the */calibration/* folder and capture an image with the See3CAM, save this as *capture_checkerboard.jpg* and run

projector_calibration.py from the terminal. This will produce *H_ctp.npy* which is a numpy array of the homography to transform the coordinate plane from the camera to the projector. Capturing the image can be done with the application “Cheese” pinned to the taskbar. Alt + Tab to the fullscreen checkerboard during the countdown. Save the image according to the earlier instructions and rotate the image and save it in the photo viewer by double clicking on the file.

5.3 Creating a New Decal Layout

Before creating a new decal layout, save the individual decals as PNGs with transparent backgrounds to */homography/decals/*. This is where the code will look to pull decals from, ensure that the names are descriptive and there are no duplicates.

Next, open *layout_spec.py*, and create a copy of the *NAPA_DECAL_LAYOUT* Python dictionary. Rename the copied dictionary to the new layout, *PENSKE_DECAL_LAYOUT* for example. Follow the same convention and replace the information with what is applicable for the new layout: decal size in inches, horizontal and vertical offset from centerline, and the pathway to the image file.



```
layout_spec.py X
.venv > Final_Pipeline_Patched > homography > layout_spec.py > NAPA_DECAL_LAYOUT

10  NAPA_DECAL_LAYOUT = {
11      "napal": {
12          "size_in": [20.5, 15.15],
13          "anchor_bottom_center_in": [NAPA_HORIZONTAL_OFFSET_IN, 16.0],
14          "image": BASE_DIR / "decals" / "napal.png",
15      },
16      "serialnumber": {
17          "size_in": [20.0, 2.1],
18          "anchor_bottom_center_in": [0.0, 1.5],
19          "image": BASE_DIR / "decals" / "serialnumber.png",
20      },
21      "usdot": {
22          "size_in": [12.25, 2.1],
23          "anchor_bottom_center_in": [0.0, 4.5],
24          "image": BASE_DIR / "decals" / "usdot.png",
25      },
26      "safetyfirst": {
27          "size_in": [12.0, 6.0],
28          "anchor_bottom_center_in": [0.0, 8.0],
29          "image": BASE_DIR / "decals" / "safetyfirst.png",
30      },
31  }
32
33  DECAL_LAYOUT = NAPA_DECAL_LAYOUT
34
```

Fig 15. Example Layout for Napa Decals

In the above example, the decal has been named *napaL* and is 20.5" x 15.15". It has a horizontal offset defined elsewhere as 0.675" and is 16" high on the centerline. Lastly, the file path to the image is given (Fig. 15).

The final step is to update the line below the layout dictionaries with the desired layout. For example, *DECAL_LAYOUT = PENSKE_DECAL_LAYOUT*, this line tells the succeeding functions which layout to use.

6. Reproduction Requirements

6.1 Flashing NVIDIA Jetson Orin Nano and Auto Boot

It is recommended to flash the Jetson before fully assembling the enclosure. For questions not covered in this manual, please refer to the Jetson [Developer Guide](#). The Jetson Orin Nano comes with firmware from the factory that does not support JetPack 6.x, which is the version of the software pack this project was built on. Therefore, the Jetson must first be flashed with [JetPack 5.1.3](#), then the firmware can be upgraded, and finally reflashed with [JetPack 6.2.1](#). On the linked pages, find “SD Card Image Method” and click the box to show the download link for the corresponding software packages.

Necessary for this step is a MicroSD card of sufficient size (at least 64GB but 128GB or higher is preferred) as well as the software [Belena Etcher](#) and [SD Card Formatter](#). Follow the [Initial Setup Guide](#) for detailed instructions on how to double check firmware and boot with the MicroSD card.

Once complete, move onto assembling the Jetson aluminum enclosure. When following the instructions for connecting the power button and pin jumper, ensure the **yellow jumper is not placed** on the auto power-on pins (5 and 6), as shown in Fig. 16). This will disable auto boot.

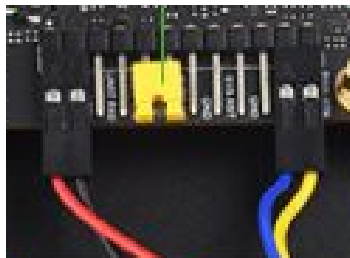


Fig 16. Do Not Place This Yellow Jumper

6.2 Startup Applications

When downloading the software pipeline from the flashdrive supplied by the UNCC Senior Design team, ensure the folder is called “Final_Pipeline” and placed in the Documents folder. Press the Windows/Super key and search for “Startup Applications”. Add a new entry

called “Start GUI” and enter this command: `/home/user/Documents/Final_Pipeline/startup.sh`. This file sends the terminal command to host the web server for the GUI.

6.3 Network Addressing

During the first time setup of the Jetson, it will prompt to connect to the internet. Once logged in, use the upper right menu to open the Wi-Fi settings. Click the settings icon next to the connected network and make note of the IPv4 address. This is the IP address that will be used to host the GUI. On a different device connected to the same network, enter `http://[IP_address]:8000`, where [IP_address] is the one noted from before, to remotely connect to the GUI once it is hosted. The startup script defines 8000 as the port number for hosting and will not change.

6.4 Necessary Driver Changes

The PL2303 driver for RS-232 serial communication is disabled by default, and must be enabled to send commands to the projector. This driver can be found in the files provided by the UNCC Senior Design team. Navigate to *SD2_FONTAINE_DECAL/Computer Code/Linux Driver* and download the *pl2303.ko*.

1. Open the file browser on the Jetson
2. Navigate to `/usr/lib/modules/5.15.148-tegra/kernel/drivers/usb/serial/`
3. Copy *pl2303.ko* into that folder
4. Replace the existing file if prompted
5. Open the terminal and run:
 - a. `sudo depmod -a`
 - b. `sudo reboot`

Once rebooted, the Jetson will use the corrected driver, enabling serial communication with the projector.

6.5 Python Setup

JetPack SDK 6.2.1 includes Python 3.10.12, so only the required packages must be installed. Installation of VS Code is optional and is used for development purposes only. Use

the terminal to navigate to the project directory `/home/user/Documents/Final_Pipeline/` and run `pip install -r requirements.txt`. Reboot the system and the GUI will automatically launch.

6.6 Aesthetic Extensions

Install the Linux extension manager to download the extensions that allow for a simplified desktop. The extension manager can be installed by running the following command in the terminal: `sudo apt install gnome-shell-extension-manager`

Open the extension manager from the application list and install *Hide Top Bar* (Author: tuor1337) and *Dash to Dock* (Author: michele_g) by searching for the extension title. Sorting by relevance or popularity may make it easier to find the correct extension. Once they are installed, return to the home page of the extension manager and make the following changes in the settings of each extension respectively:

Hide Top Bar > *Intellihide* > *Disable "Only hide panel when a window takes the space"*

Dash to Dock > *Intelligent autohide* > *Additional Settings button* > *Disable "Dodge windows"*

In the Downloads folder of the original Jetson provided by the UNCC Senior Design team, there are light and dark versions of the Fontaine wallpaper used, which can be reused on reproduced systems.

7. Maintenance Plan

Before and after every setup, it is necessary to inspect the main power cable for damage to prevent electrical shock. Electrical shock can cause serious injury to the operator and damage system components. Be careful when moving the system to not roll over the main power cable.

As the system is moved, it is possible for the caster wheels to begin unthreading if they are not tightened properly. Check the caster wheels on a weekly basis to ensure they remain tight. If the system ever feels unstable, stop operation immediately and check the caster wheels. Ensure that all four wheels contact the ground when locking so that the system doesn't move during use.

Clean the lens of the projector and camera as needed. Use of the included camera lens cover is highly recommended when the system is not in operation. If the focus of the projector lens is affected during cleaning, rotate the outer focus ring such that the silver marks align. These marks are to ensure repeatable focus at 8ft.